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Professional Summary

Accomplished Machine Learning Engineer with 9+ years of expertise in AI/ML, deep learning, Generative AI (LLMs, RAG, NLP), and MLOps. Specializes in designing, deploying, and optimizing AI solutions at scale, integrating advanced ML models into production environments. Proficient in Python, R, and JavaScript, with expertise in cloud platforms (AWS, Azure, GCP) and big data technologies (Hadoop, Spark, Kafka). Skilled in ML model development, deployment, versioning, and monitoring using CI/CD, Kubernetes, and Docker. Adept at predictive analytics, TensorFlow, PyTorch, Scikit-learn, and XGBoost, with a focus on building scalable data pipelines and automating ML workflows. Experienced in integrating Generative AI and LLMs into enterprise applications, enhancing AI-driven decision-making and operational efficiency.

Technical Skills

- **Programming Languages:** Python, R, C, C++, JavaScript, PySpark, ASP.NET
- **Technical Skills:** HTML, CSS, React, Angular, Neo4j (graph database)
- **Machine Learning & Analytics:** Scikit-learn, TensorFlow, PyTorch, XGBoost, NLTK, Automated Speech Recognition (ASR), NLU, NLP, Google AutoML, Microsoft Azure AutoML, Tableau, Reinforcement Learning
- **Big Data Technologies:** Hadoop, Spark, Kafka, Cloudera, Cassandra, HDFS, MapReduce, Apache HBase, Google BigQuery, Hive
- **Cloud Platforms:** AWS, Azure, Google Cloud Platform (GCP), VertexAI, OpenAI, ChatGPT.
- **Data Handling & Visualization:** SQL, PL/SQL, Power BI, SSRS, SSAS, MySQL, MongoDB, Matplotlib, Seaborn, NoSQL (Cassandra), AWS S3, Google Cloud Storage.
- **Tools & Technologies:** Sqoop, Git, Docker, Kubernetes, Apache Airflow, Jenkins, Elasticsearch, Kibana (ELK Stack), Grafana, Prometheus.
- **MLOps & Automation:** AWS SageMaker, Azure ML, Google Cloud Platform (GCP), Vertex AI, AutoML, CI/CD Pipelines, Kubernetes, Kubeflow, GCP IAM, Terraform, Model Versioning, MLflow, Model Management (MLOps).

Education

Master's Degree in Computer Science, University of North Texas

Bachelor's Degree in Computer Application, Sardar Patel University

Professional Experience

Role: Machine Learning Engineer | Client: Nike | Remote | Atlanta, GA

Aug 2023 – Present

- Configured and optimized the development environment by setting up Python and installing essential ML libraries (TensorFlow, PyTorch, Scikit-learn, Spark ML), ensuring compatibility and performance for model development, training, and deployment.
- Leveraged Databricks and big data technologies such as Spark on GCP Dataproc, Hadoop, Kafka, and Scala for large dataset processing, while managing databases (SQL, MySQL) for structured data storage.
- Developed, trained, and fine-tuned ML models for text classification (15% accuracy improvement) and NLP relevance tuning using TensorFlow, Scikit-learn, and Spark ML, deploying models on GCP Vertex AI with mlFlow and automated monitoring pipelines.
- Integrated AWS SageMaker and GCP Vertex AI for scalable model training and deployment, leveraging cloud infrastructure to enhance performance, reduce latency, and ensure seamless model lifecycle management. Conducted A/B testing and cross-validation to improve accuracy, applying evaluation metrics (precision, recall, F1-score) for robustness. Automated ETL pipelines for conversational AI, reducing runtime by 30% with seamless GCP Vertex AI integration.

- Designed, developed, and integrated RAG pipelines using Elasticsearch for vector database optimization (25% search relevance increase), utilized GCP AI tools (Vertex AI, AutoML, Seldon) and GKE for scalable deployment, developed ASR and NLU models for Conversational AI, implemented CI/CD pipelines with Git, Docker, Kubernetes, managed model versioning, and designed IaC solutions with Terraform for efficient ML model training and deployment management.
- Set up automated monitoring and logging systems to track model performance, trigger alerts for anomalies, and schedule retraining workflows, ensuring continuous reliability in production. Designed and implemented advanced conversational AI solutions using transformer models (GPT, BERT), RAG pipelines, and Neo4j for recommendation engines, optimized search relevance models with Elasticsearch and vector embeddings (20% user engagement increase), and developed Tableau dashboards for real-time insights and data-driven decision-making.

Role: Machine Learning Engineer | Client: Goldman Sachs | Remote | Mumbai, India **Jul 2020 – Dec 2021**

- Led end-to-end analytics projects leveraging Python, encompassing data preprocessing, feature engineering, ML model development, training, and deployment. Utilized AWS SageMaker and Docker for scalable deployment, while implementing monitoring and maintenance strategies using tools such as Scikit-learn and TensorFlow to ensure model performance and reliability in production.
- Conducted statistical analysis for rate-making and customer insights, increasing forecasting accuracy by 20%, using ML algorithms and text analytics. Designed and implemented generative AI solutions, leveraging LLMs for text clustering, topic modeling, and retrieval-augmented workflows. Additionally, enhanced search relevance with behavioral analytics and OpenAI services like ChatGPT.
- Utilized Sqoop for data transformation and automated CI/CD pipelines for model deployment using AWS SageMaker and AWS Step Functions, streamlining workflows and integrating ML lifecycle stages with enhanced scalability and error handling.
- Maintained and monitored deployed models for performance drift, retraining, and optimization, ensuring scalability and reliability in production. Designed and implemented cognitive AI algorithms leveraging classical AI techniques like expert systems and first-order logic for decision support systems.
- Developed solutions for automated speech recognition (ASR) and NLU models, improving data interaction and customer experience.
- Generative AI Solutions & Customer Insights: Developed and deployed generative AI models and NLP-based analytics to extract customer insights, optimize financial decision-making, and integrate OpenAI-powered AI assistants into enterprise applications for automated reporting and intelligence.
- Designed and implemented generative AI solutions to personalize customer interactions, leveraging LLMs for content creation, recommendations, and dynamic customer support, driving a 30% increase in engagement. Leveraged behavioral analytics and generative AI to enhance customer recommendation systems, increasing conversion rates by 25%. Integrated conversational AI for sentiment analysis and predictive analytics, delivering real-time insights that improved customer satisfaction and retention.

Role: Machine Learning Engineer | Client: UnitedHealth Group | Mumbai, India **Feb 2018 – Jun 2020**

- Deployed production-grade machine learning (ML) models on a big data platform, leveraging Python, React, Scala, Hive, and Hadoop for scalable data processing. Utilized Azure cloud architecture and Azure ML for seamless model deployment, ensuring efficient integration and performance optimization in a production environment.
- Developed efficient big data solutions using Hadoop and web services (WCF, .Net) on Azure. Enhanced ML algorithm accuracy by 12% through optimized data preprocessing, robust unit testing (Python), and iterative model refinement. Optimized model performance and scalability, achieving a 30% increase in throughput, by leveraging Spark ML in Scala and Azure cloud resources.
- Proficient in ML lifecycle stages, including data preprocessing, model creation, training, development, deployment, monitoring, and maintenance, leveraging Azure ML for scalable solutions

